

HS-401: Professional Values and Ethics

LECTURE NO 3: *Codes of Conduct*

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1.1 Introduction (1/3)

Case Bay Area Rapid Transport Project



Figure 2.1 BART train. Photo: NISEE-PEER Earthquake Engineering Library, University of California, Berkeley.

In March 1972 Holger Hsортsvang, Max Blakenzee, and Robert Bruder, three engineers, working on the *Bay Area Rapid Transport Project (BART)* in California (United States) and responsible for the design and creation of an automatic guided train system, were dismissed. These engineers had been expressing their doubts about the safety of the system via internal memos since 1969 to their managers. The response was “don’t make trouble.” In 1971 they brought their concerns in confidence to members of the board of directors, thus bypassing their immediate superiors. That was unconventional

1.1 Introduction (2/3)

for the BART organization and indeed for any hierarchical organization. The director they finally made contact with turned out to be very interested in their case and so he promised to raise it with the management. He furthermore promised to keep their names anonymous and do nothing to damage their interests. However, two days after the encounter the full story was published in the *Contra Costa Times*. At first the engineers denied having any involvement in the matter but once their involvement was confirmed they were immediately fired without cause or appeal. They subsequently took the matter to court.

In the wake of the affair one of the organizations to become involved was the Institute of Electrical and Electronic Engineers (IEEE). The IEEE decided to send what is known as an amicus curiae letter to the law courts. (An amicus curiae is an “friend of the court”: someone, not a party to a case, who voluntarily offers information on a point of law or some other aspect of the case to assist the court.) The letter emphasized the fact that according to the IEEE’s professional code, engineers are responsible for the “safety, health and welfare of the public.” The IEEE also argued that the professional code is an implicit aspect of the employment contract. If this argument had been accepted by the judge then it would have meant that employees who act in accordance with what is stated in the professional code may not be simply dismissed.

1.1 Introduction (3/3)

After the three engineers had lost their job, their concerns were decisively confirmed on October 2, 1972, three weeks after BART began carrying passengers. There was a train system accident and several passengers were injured. Despite this, the three engineers accepted an out-of-court settlement reported to be \$25,000 per person. The presumed reason for this was that they had in the first instance lied about their involvement in the matter which had weakened their case. Apart from anything else, the dismissals were very detrimental for the careers of all three engineers.

Source: Based on Anderson, Perucci, Schendel, and Trachtman (1980), Anderson, Otten, and Schendel (1983), and Unger (1994, pp. 12–17).

- In this case, the three engineers acted out of a sense of professional responsibility. This professional responsibility was codified in the IEEE code of conduct and was related to the safety, health, and welfare of the public. Although their professional organization supported their behaviour, it could not prevent them from being dismissed. In this lecture, we will discuss the role of codes of conduct in engineering. In particular, we focus on *professional codes* as they have been proposed by professional engineering societies and on *corporate codes*, as they have been formulated by companies.

1.2 Codes of Conduct (1/2)

- ***Codes of conduct:*** A code in which organizations (like companies or professional associations) lay down guidelines for responsible behavior of their members.
- Codes of conduct are often intended as an addition to the requirements of the law. When codes of conduct are enforced this is usually done by the organization that formulated the code.
- For engineers, two types of codes of conduct are especially important:
 - ***Professional code-*** Code of conduct that is formulated by a professional association.
 - ***Corporate code-*** Code of conduct that is formulated by a company.

1.2 Codes of Conduct (2/2)

- Depending on the exact objectives of a code of conduct, a distinction can be made between three types of codes of conduct:
 - *Aspirational code*- A code that expresses the moral values of a profession or company. The objective of such a code is to express to the outside world the kind of values the profession or company is committed to.
 - *Advisory code*- A code of conduct that has the objective to help individual professionals or employees to exercise moral judgments in concrete situations.
 - *Disciplinary code*- A code that has the objective to achieve that the behavior of all professionals or employees meets certain values and norms.

1.2.1 Professional Codes (1 / 2)

- Professional codes are guidelines for the exercising of a profession and are formulated by a professional society.
- Professional codes have been formulated for a variety of professions like doctors, nurses, lawyers, priests, the police, and corporate managers etc.
- Also engineers have professional codes of conduct.

1.2.1 Professional Codes (2/2)

- All professional codes include the obligation to practice one's profession with *integrity* and *honesty*, and in a competent way.
- With integrity and honesty we mean that the profession must be conducted in an honest, faithful, and truthful manner.
- Sometimes it is also stipulated that the profession must be practiced in an independent and impartial way. Usually this is meant to imply that engineers should avoid *conflicts of interests*. You have a conflict of interest if you have an interest that, when pursued conflicts with meeting your obligations to your employer or clients.

1.2.2 Corporate Codes (1/5)

- Corporate codes are voluntarily commitments made by individual companies or associations of companies setting certain values, standards, and principles for the conduct of corporations.
- The formulation of corporate codes is based on the assumption that companies have a *corporate social responsibility*, that is, a responsibility towards stakeholders and to society at large.
- We will discuss the main elements of the various kinds of corporate codes: *the mission*, the *core values*, the *responsibilities towards stakeholders* and detailed *rules and norms*.

1.2.2 Corporate Codes (2/5)

Mission statement

- Many corporate codes contain a mission statement that concisely formulates the strategic objectives of the company and answers the question what the organization stands for.
 - *WhatsApp started as an alternative to SMS. Our product now supports sending and receiving a variety of media: text, photos, videos, documents, and location, as well as voice calls. Some of your most personal moments are shared with WhatsApp, which is why we built end-to-end encryption into our app. Behind every product decision is our desire to let people communicate anywhere in the world without barriers.(WhatsApp Mission Statement).*

1.2.2 Corporate Codes (3/5)

Core values

- Core values express the qualities that a company considers desirable.
- As a company, and as individuals, we value:
 - Integrity and honesty.
 - Passion for customers, for our partners, and for technology.
 - Openness and respectfulness.
 - Taking on big challenges and seeing them through.
 - Constructive self-criticism, self-improvement, and personal excellence.
 - Accountability to customers, shareholders, partners, and employees for commitments, results, and quality. (*Microsoft*)

1.2.2 Corporate Codes (4/5)

Responsibility towards stakeholders

- Most corporate codes also express responsibilities to a variety of stakeholders like consumers, employees, investors, society, and the environment. Competitors and suppliers are also sometimes mentioned as stakeholders.
- In addition to responsibilities towards stakeholders, some corporate codes also contain *stakeholder principles* that guide the relationship between company and its stakeholders.

1.2.2 Corporate Codes (5/5)

Norms and rules

- Norms and rules contain guidelines for employees how to act in specific situations. This may include subjects like the acceptance of gifts, fraud, conflicts of interest, confidentiality, theft, corruption, bribery, discrimination, respect, and sexual harassment.
- Some rules from *Intel's* How the Corporate Business Principles Apply to You:
 - Employees must follow the law wherever they are around the world and in all circumstances. Do not engage in behavior that harms the reputation of Intel or yourself.
 - Employees must avoid both actual and perceived conflicts of interest.
 - Customers and suppliers must be dealt with fairly and at arm's length.
 - Employees must never attempt to bribe or improperly influence a government official, customer or supplier.

1.3 Possibilities and Limitations of Codes of Conduct

- As we have seen, codes of conduct help to express the responsibilities of engineers. Still, in the course of time, a number of objections against code of conduct have been levelled. In judging these objections, one should keep in mind that codes of conduct may have different objectives. Especially the difference between aspirational, advisory, and disciplinary codes is relevant here.

1.3.1 Codes of conduct and self-interest (1/2)

- Codes of conduct are a form of self-regulation. Sometimes, they are primarily formulated for reasons of self-interest, for example to improve one's image to the outside world, to avoid government regulation or to silence dissident voices. An example in which the latter happened is the case of Jon Tozer .

Case John Tozer

In 1989 the Australian engineer John Tozer criticized the decision of the Coffs Harbour authorities to pump sewage into the sea. According to him the engineers employed by the local authority had given a misleading impression of the effects upon the environment and they had failed to properly investigate the alternatives. The engineers in question were subsequently successful in removing Tozer from the Association of Consulting Engineers Australia (ACEA). Tozer was accused of having contravened the professional code by openly criticizing the work of other (associated) engineers. Because of his disbarment Tozer, who has his own consulting engineering firm, is no longer able to fulfill any contracts for customers demanding ACEA membership.

Source: Based on Beder (1993).

1.3.1 Codes of conduct and self-interest (2/2)

- The fact that self-interest plays a role in formulating codes of conduct is not necessarily objectionable as long as the content of the code is ethical. One way to ensure this is to include a range of stakeholders in the formulation and implementation of the code of conduct to avoid the code becoming one-sided.

1.3.2 Vagueness and potential contradictions

- In the application of codes of conduct to concrete situations, one is frequently confronted with rather vague concepts and rules that need interpretation. Depending on the exact interpretation of such concepts and rules, codes of conduct sometimes result in contradictory recommendations about what to do in a specific situation.
- One relevant notion from codes of conduct that is in need of further clarification and interpretation is “loyalty.”
 - *Critical loyalty*- implies that an employee has a right (and responsibility) to (internally) criticise actions by the employer when there is a sincere belief that the action is detrimental to the interest of the employer, or harmful to others (employees or public).
 - *Uncritical loyalty*- implies that the employee supports (actively or passively) all actions by the employer.

1.3.3 Can ethics be codified?

- Some authors have argued that the idea of drafting a code of conduct is misperceived because ethics cannot be codified.
- Whereas people who criticize the vagueness and potential contradictions in codes of conduct are worried that such codes do not uniformly prescribe certain behavior, people who argue that ethics cannot be codified are often worried that codes of conduct contain strict prescriptions which conflict with what ethics is about according to them.
- One needs to accept some degree of vagueness and some potential conflicts in codes of conduct.

1.3.4 Can codes of conduct be lived by?

- Codes of conduct sometimes contain provisions that are very difficult or impossible to follow in practice.
- Professional codes sometimes require that engineers inform the public timely and completely if the safety, health, or welfare of the public is put at stake in a technological project. This duty to inform the public can conflict with the confidentiality of the duty that engineers also have according to the law in many countries. If engineers in such situations release information outside the company in which they are working, they are blowing the whistle.
- Engineers, and other employees, who blow the whistle are usually in a weak position from a legal point of view. The situation is different from country to country, but the laws that regulate employment contracts in most countries either impose certain *confidentiality duties* on employees or they allow the employer to order the employee to keep silent about certain specific information, or they do both.

1.3.5 Enforcement

- Enforcement is only an objective in the case of disciplinary codes. Active enforcement of codes of conduct seems to be an exception, especially for professional codes.

1.4 Codes of Conduct in an International Context

1.4.1 Global codes for multinationals (1/2)

- *Global code of conduct*- A code of conduct that is believed to apply worldwide.

United Nations Global Compact Principles

Human Rights

Principle 1: Businesses should support and respect the protection of internationally proclaimed human rights.

Principle 2: Make sure that they are not complicit in human rights abuses.

Labour Standards

Principle 3: Businesses should uphold the freedom of association and the effective recognition of the right to collective bargaining.

Principle 4: The elimination of all forms of forced and compulsory labour.

Principle 5: The effective abolition of child labour.

Principle 6: The elimination of discrimination in respect of employment and occupation.

1.4.1 Global codes for multinationals (2/2)

United Nations Global Compact Principles

Environment

Principle 7: Businesses should support a precautionary approach to environmental challenges.

Principle 8: undertake initiatives to promote greater environmental responsibility.

Principle 9: encourage the development and diffusion of environmentally friendly technologies.

Anti-Corruption

Principle 10: Businesses should work against corruption in all its forms, including extortion and bribery.

1.4.2 Global codes for engineers (1/3)

- The globalization of the world's economies has also increased the working space of engineers. Engineering products and production facilities often transcend national boundaries. Engineers travel across the world and meet other cultures by interacting with foreign engineers. Multinational companies employ engineers from different cultural backgrounds in the same corporate environment. So, engineering has become a global activity and increasingly requires a global approach and acceptable global guidance.
- The engineering profession in the United States has been a world leader in promoting engineering ethics code development and associated educational activities.
- US codes are based on the notion of *professional autonomy*: “empowering individuals to reason more clearly and carefully concerning moral questions, rather than to inculcate any particular beliefs”

1.4.2 Global codes for engineers (2/3)

Ethical Principles for Engineers in a Global Environment

- *The Principle of Public Safety:* Engineers should endeavour, based on their expertise, to keep members of the public safe from serious negative physical consequences resulting from their development and implementation of technology.
- *The Principle of Human Rights:* Engineers should endeavour to ensure that fundamental rights of human beings will not be negatively impacted as a result of their work with technology.
- *The Principle of Environment and Animal Preservation:* Engineers should endeavour to avoid damage to the animal kingdom and the natural environment which would result in serious negative consequences, including long-term ones, to human life.
- *The Principle of Engineering Competence:* Engineers should endeavour to engage only in engineering activities which they are competent to carry out.

1.4.2 Global codes for engineers (3/3)

Ethical Principles for Engineers in a Global Environment

- *The Principle of Scientifically Founded Judgment:* Engineers should endeavour to base their engineering decisions on scientific principles and mathematical analysis, and seek to avoid influence of extraneous factors.
- *The Principle of Openness and Honesty:* Engineers should endeavour to keep the public informed of their decisions which have the potential to seriously affect the public, and to be truthful and complete in their disclosures. (Luegenbiehl, 2010)

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